EE426 / 626 Microprocessor Applications

Brief History of the Intel 4040 → 8008 → 8080



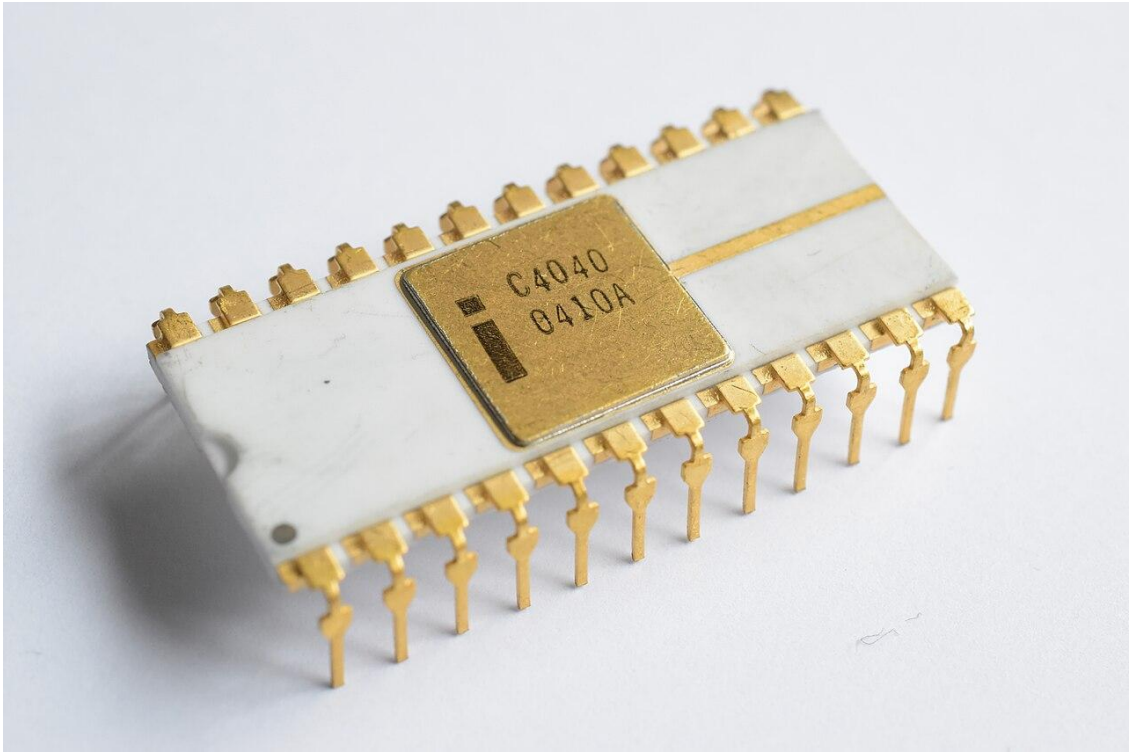
EE426 / 626 Microprocessor Applications

Intel 4004 → 4040

MCS-4

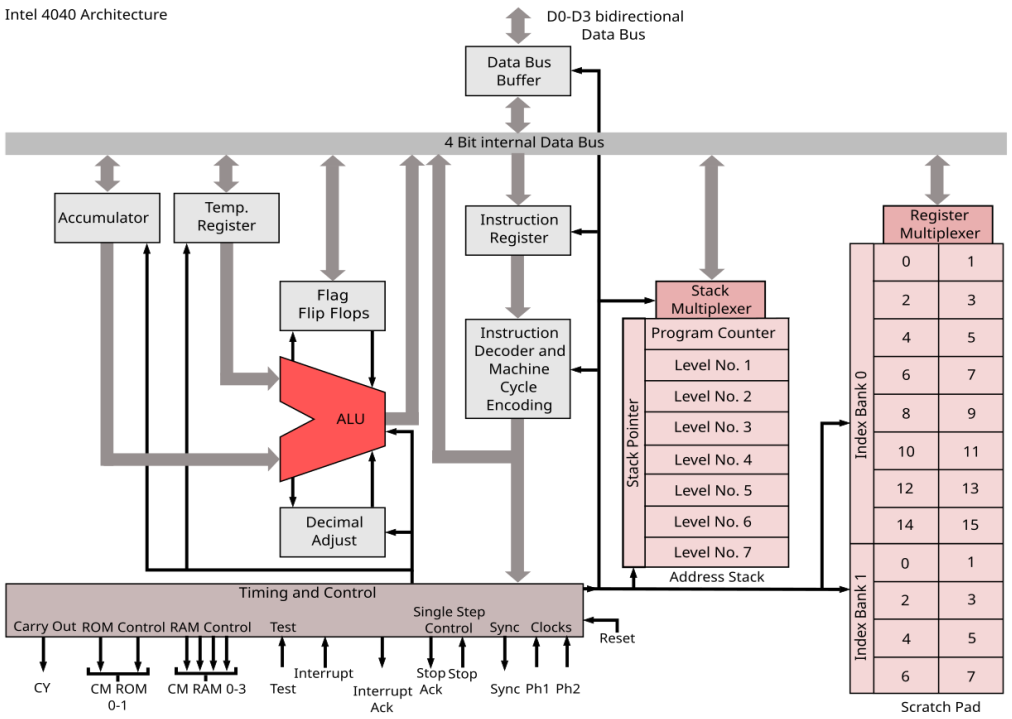
1974 Intel releases 4040 (successor to 4004)

- 10 μm process using PMOS
- 24-pin PDIP/CDIP
- 12-bit address + 1-pin extend (8 k 4-bit ROM)
- −15 VDC Supply
- 500 kHz to 740 kHz clock
- Interrupt support (missing on 4004)
- 7-level stack
- 60 instructions (up from 46)
- 24 index registers, banked for contect switching
  - Bank 0: R0–R15; Bank 1: R16–R23, bank-switched, R8–R15 shared)
- Design Lead: Federico Faggin

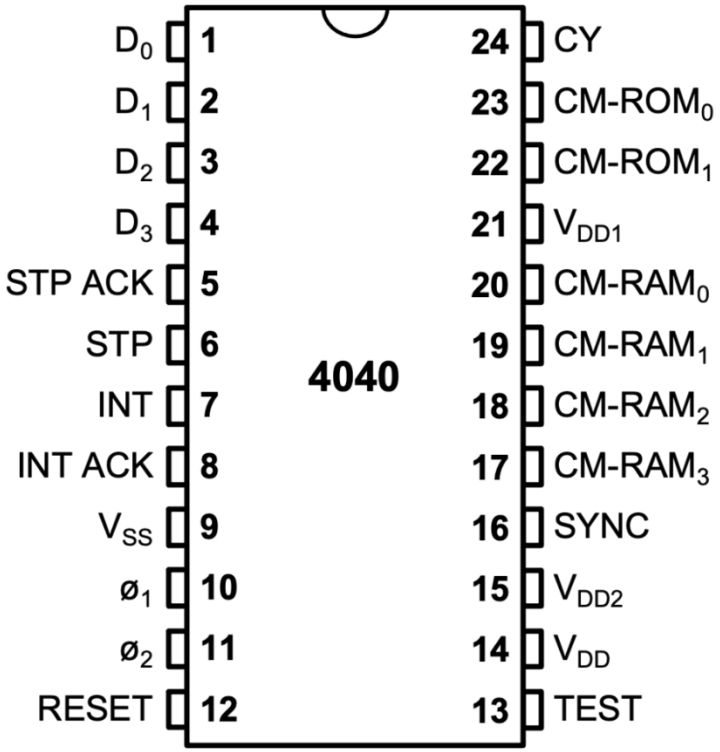


By Stelo.xyz, Pttn, or Thomas Nguyen, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=49507562

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intel 4040

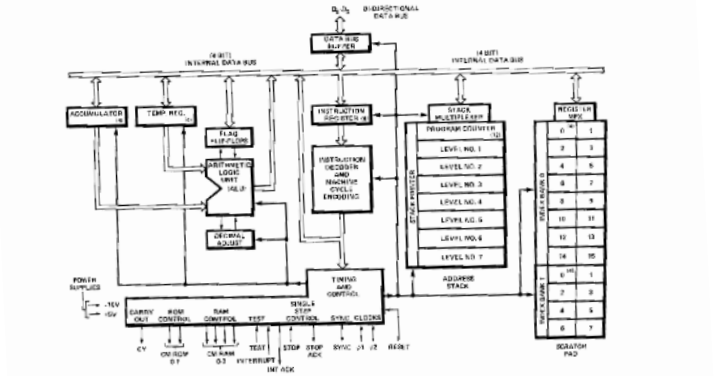
SINGLE CHIP 4-BIT P-CHANNEL MICROPROCESSOR

- Functionally and Electrically Upward Compatible to 4004 CPU
- 14 Additional Instructions (60 total) Including Logical Operations and Read Program Memory
- Interrupt Capability
- Single Step Operation
- 8K Byte Memory Addressing Capability
- 24 Index Registers
- Subroutine Nesting to 7 Levels
- Standard Operating Temperature Range of 0° to 70° C
- Also Available With -40° to +85° C Operating Range

The Intel® 4040 is a complete 4-bit parallel central processing unit (CPU). The CPU can directly address 4K eight bit instruction words or 8K with a bank switch. Seven levels of subroutine nesting, including interrupt, and 24 randomly accessable index registers (24x4) are provided as convenient facilities for the designer. The index registers may be used for addressing or for scratch pad memory for storing computation results. The interrupt feature permits a normal program sequence to be interrupted, with normal program execution continuing after the interrupt service routine is completed. Provisions have also been made to permit single-stepping the CPU using the STOP and ACKNOWLEDGE signals.

The 4040 is an enhanced version of the 4004 and as such retains all the functional capability of that device. It will execute all the 4004 instructions, and is also electrically compatible with all components used with a 4004 CPU.

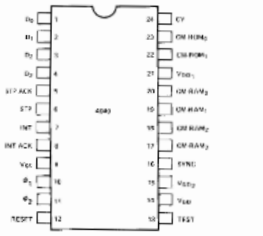
BLOCK DIAGRAM



Fair use, Intel 4040 Datasheet

4040

Pin Description



D_0-D_3
BIDIRECTIONAL DATA BUS. All address and data communication between the processor and the RAM and ROM chips occurs on these 4 lines.

STP
STOP input. A logic "1" level on this input causes the processor to enter the STOP mode.

STPA
STOP ACKNOWLEDGE output. This signal is present when the processor is in the stopped state. Output is "open drain" requiring pull-down resistor to V_{DD}.

INT
INTERRUPT input. A logic "1" level at this input causes the processor to enter the INTERRUPT mode.

INTA
INTERRUPT ACKNOWLEDGE output. This signal acknowledges receipt of an INTERRUPT signal and prevents additional INTERRUPTS from entering the processor. It remains active until cleared by the execution of the new BRANCH BACK and SRC (BBS) instruction. The output is "open drain" requiring a pull-down resistor to V_{DD}.

RESET
RESET input. A logic "1" level at this input clears all flag and status registers and forces the program counter to zero. To completely clear all address and index registers, RESET must be applied for 96 clock cycles (12 machine cycles).

TEST
TEST input. The logical state of this signal may be tested with the JCN instruction.

SYNC
SYNC output. Synchronization signal generated by the processor and sent to ROM and RAM chips. It indicates the beginning of an instruction cycle.

$CM\text{-}RAM_0$ – $CM\text{-}RAM_3$
CM-RAM outputs. These are bank selection signals for the 4002 RAM chips in the system.

$CM\text{-}ROM_0$ – $CM\text{-}ROM_1$
CM-ROM outputs. These are bank selection signals for program ROM chips in the system.

CY
CARRY output. The state of the carry flip flop is present on this output and updated each X_1 time. Output is "open drain" requiring pull down resistor to V_{DD}.

ϕ_1, ϕ_2	Two phase clock inputs
V_{SS}	Most positive voltage
V_{DD}	V_{SS}–15V ±5% – Main supply voltage
*V_{DD_1}	V_{SS}–15V ±5% – Timing supply voltage
**V_{DD_2}	– Output buffer supply voltage

*For low power operation
**May vary depending on system interface

Fair use, Intel 4040 Datasheet

EE426 / 626 Microprocessor Applications



MICRO 440... the MORE affordable computer!

The MICRO 440, with a price intended to turn the competition's hair gray. It's built around the Intel 4040 CPU. And although it has only a 4-bit word, it's no toy. It features a powerful set of 60 instructions to permit binary and decimal arithmetic, logic, loops and branching and real-time interrupts. It also features 24 on-chip registers (more than many more expensive micros) and subroutine nesting to seven levels. With the bare-bones kit you get a **single** PC board housing CPU, clock, I/O interface, front panel controls and displays, and 256 bytes of program/data RAM. You also get our complete and instructive 87-page manual. Order the complete kit, and you also get an enclosure, power supply, a **standard** input and output port and a **standard** Teletype interface. Need more memory? There's room for 8K of RAM or PROM memory, in 2K units. Each memory unit gives you extra I/O ports.

The age of the microprocessor is here. They're already in sewing machines, parking garages, even pin-ball machines. Soon they'll be cooking your meals and controlling your TV. Have you been wanting to learn more about this exciting new technology? You could spend $300 to $600 to attend a course. Or, for less money, you can get unlimited hands-on experience with your own MICRO 440. With the basic kit you can enter a program up to 256 steps long, run and display the results. Following the simple instructions in our user's manual, you will learn the workings of the computer by loading and executing the many sample programs given, then learn to write your own programs. The usefulness of the MICRO 440 doesn't vanish as you learn. It's powerful enough to grow with you. Plug a Teletype or TV Typewriter into the back panel and greatly expand its capabilities. Add memory and peripherals, and you're ready for serious, efficient computing.

Even fully expanded, the MICRO 440 won't cost you an arm and a leg. Would you believe a computer with 2K PROM, 6K RAM, 32 I/O ports, TTY interface, **and** a TV Typewriter for less than $1300?

PRICES: Bare Bones Kit **$175**
Complete Kit $275
Assembled & Tested ... $375

PARTIAL KITS ALSO AVAILABLE.
WRITE FOR PRICES.

COMP- SULTANTS, Inc.

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HUNTSVILLE, ALABAMA 35807
205-837-5100

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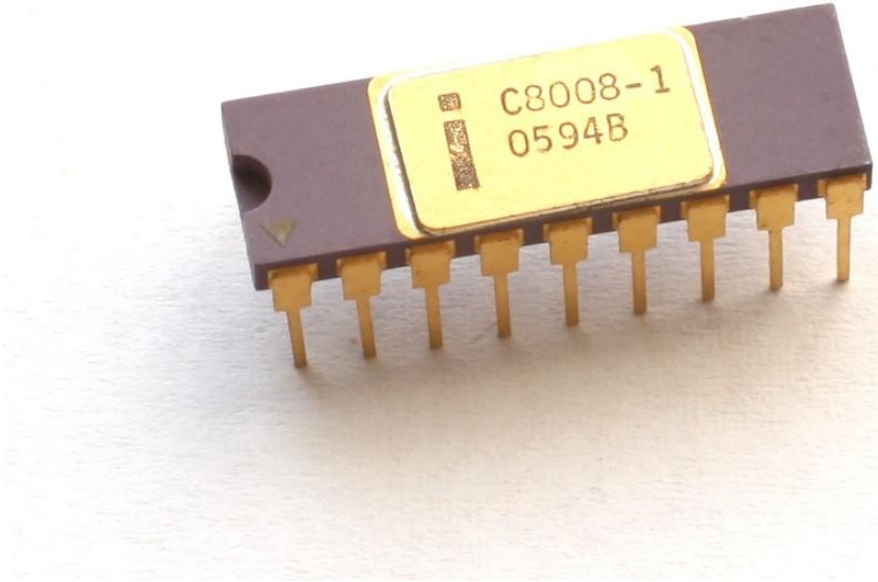
EE426 / 626 Microprocessor Applications

1201 (aka Intel 8008 and TI 1795)

MCS-8

1972 Intel releases 8008 (8-bit processor)

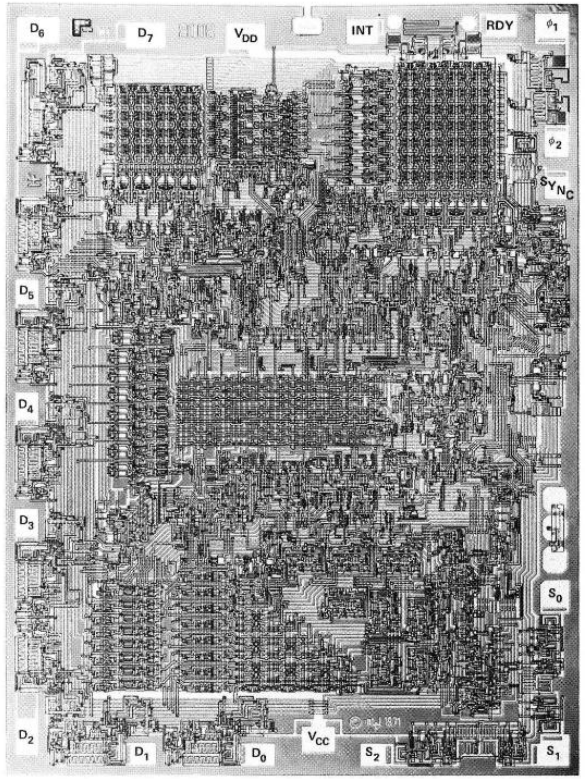
- Designed by Computer Terminal Corp (CTC)
- 10 μm process using PMOS
- 18-pin PDIP/CDIP
- 14-bit address (16 kB RAM / ROM)
- +5 V_{CC} Supply, –9 V_{DD} Supply
- 500 kHz clock
- Interrupt support, 7-level stack
- 48 instructions
- Design Lead: Federico Faggin
- Architecture Design: Victor Poor (CTC)
- INTERP/8 Simulator (in FORTRAN IV, Gary Kildall)



By Konstantin Lanzet - CPU Collection Konstantin LanzetCamera: Canon EOS 400D, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=5694177

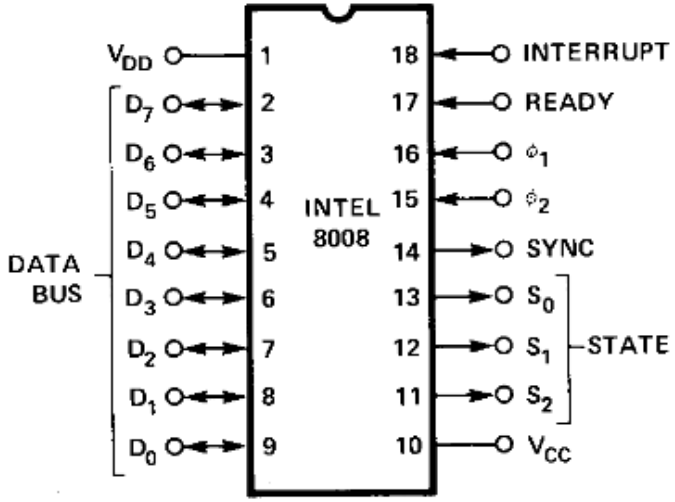
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Intel 8008 Die



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PIN CONFIGURATION



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EE426 / 626 Microprocessor Applications

CTC Datapoint 2200



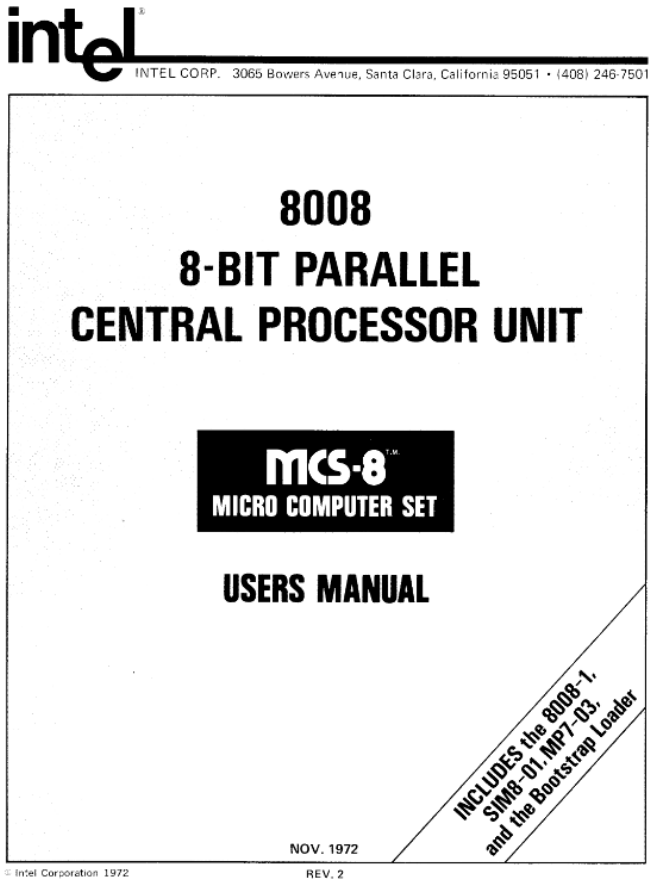
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Intel 8008 Register Set

$^{1}_{3}$ $^{1}_{2}$ $^{1}_{1}$ $^{1}_{0}$ $^{0}_{9}$ $^{0}_{8}$ $^{0}_{7}$ $^{0}_{6}$ $^{0}_{5}$ $^{0}_{4}$ $^{0}_{3}$ $^{0}_{2}$ $^{0}_{1}$ $^{0}_{0}$ *(bit position)*		
Main registers		
	A	**A**ccumulator
	B	**B** register
	C	**C** register
	D	**D** register
	E	**E** register
	H	**H** register *(indirect)*
	L	**L** register *(indirect)*
Program counter		
PC		**P**rogram **C**ounter
Push-down address call stack		
AS		Call level 1
AS		Call level 2
AS		Call level 3
AS		Call level 4
AS		Call level 5
AS		Call level 6
AS		Call level 7
Flags		
C P Z S		Flags[13][a]

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intel

INTEL CORP. 3065 Bowers Avenue, Santa Clara, California 95051 • (408) 246-7501

8008

8-BIT PARALLEL CENTRAL PROCESSOR UNIT

MCS-8™ MICRO COMPUTER SET

USERS MANUAL

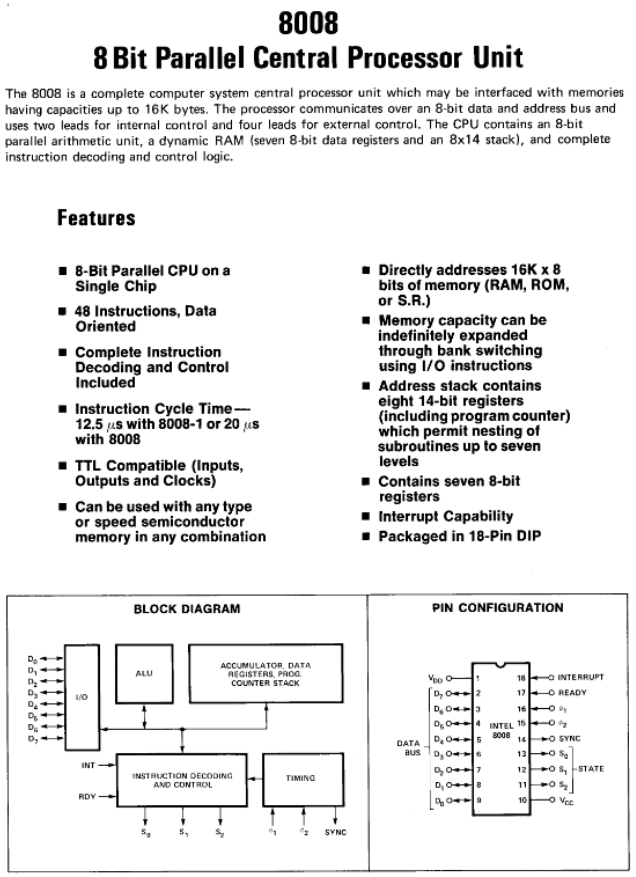
INCLUDES the 8008-1, SIM8-01, MP7-03, and the Bootstrap Loader

NOV. 1972

REV. 2

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8008

8 Bit Parallel Central Processor Unit

The 8008 is a complete computer system central processor unit which may be interfaced with memories having capacities up to 16K bytes. The processor communicates over an 8-bit data and address bus and uses two leads for internal control and four leads for external control. The CPU contains an 8-bit parallel arithmetic unit, a dynamic RAM (seven 8-bit data registers and an 8x14 stack), and complete instruction decoding and control logic.

Features

- 8-Bit Parallel CPU on a Single Chip
- 48 Instructions, Data Oriented
- Complete Instruction Decoding and Control Included
- Instruction Cycle Time— 12.5 μs with 8008-1 or 20 μs with 8008
- TTL Compatible (Inputs, Outputs and Clocks)
- Can be used with any type or speed semiconductor memory in any combination
- Directly addresses 16K x 8 bits of memory (RAM, ROM, or S.R.)
- Memory capacity can be indefinitely expanded through bank switching using I/O instructions
- Address stack contains eight 14-bit registers (including program counter) which permit nesting of subroutines up to seven levels
- Contains seven 8-bit registers
- Interrupt Capability
- Packaged in 18-Pin DIP

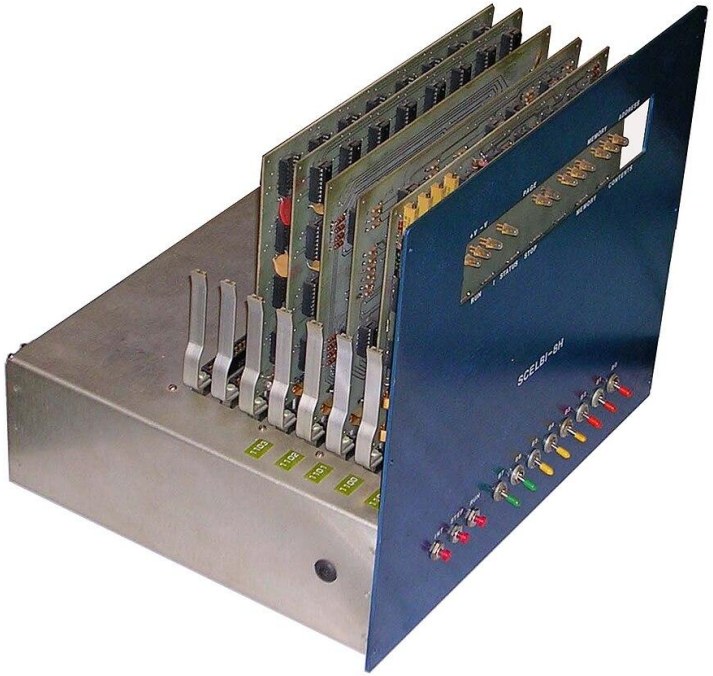
BLOCK DIAGRAM

PIN CONFIGURATION

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SCELBI 8H Computer (Kit)



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MCM/70 Microcomputer



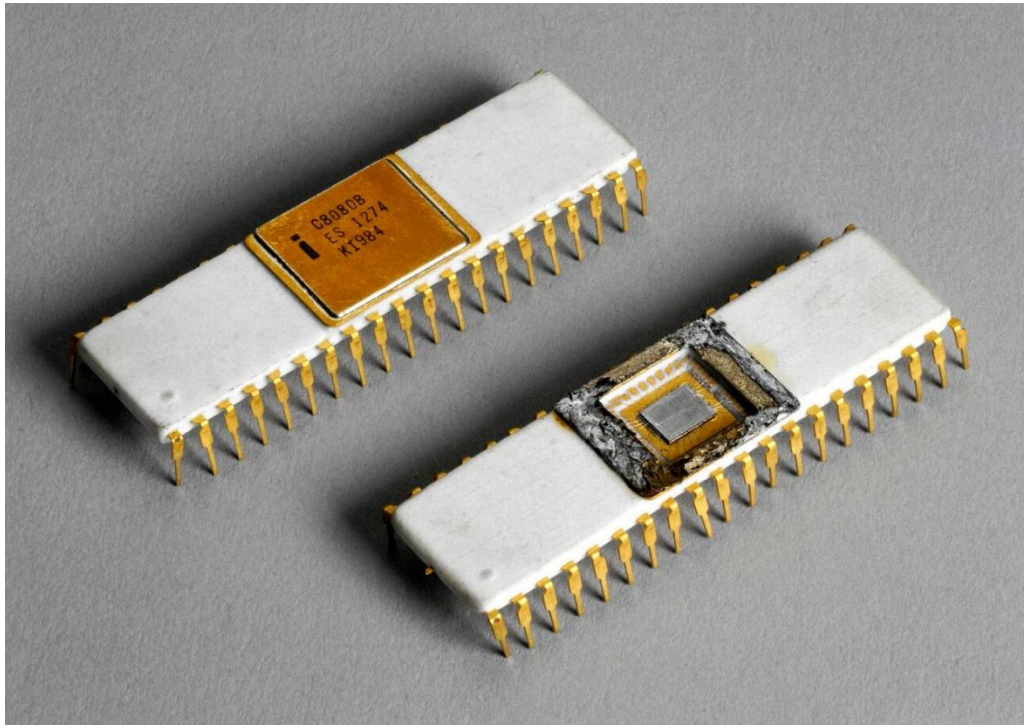
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EE426 / 626 Microprocessor Applications

Intel 8080 / 8080A

MCS-80
1974 Intel releases 8080 (8-bit 8008 successor)

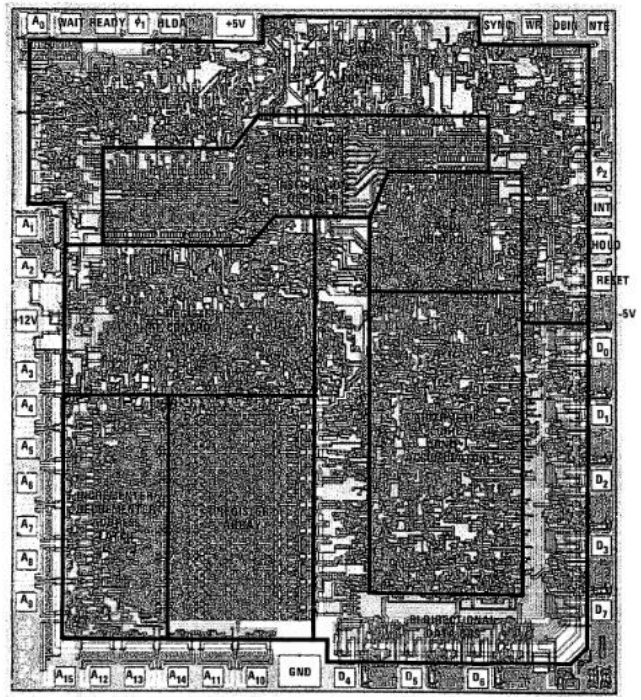
- Intel Design
- 10 μm process using NMOS TTL Compatible
- 40-pin PDIP/CDIP
- 16-bit address (64 kB RAM / ROM)
- +12 V_{DD}, +5 V_{CC}, −5 V_{BB} Supplies
- 2 MHz to 2.63 MHz clock speeds
- Interrupt support, External stack
- 78 instructions
- Design Architect: Federico Faggin
- Lead Design: Masatoshi Shima
- INTERP/80 Simulator (in FORTRAN IV by Gary Kildall)



By The Science Museum UK - [1], CC BY 4.0, https://commons.wikimedia.org/w/index.php?curid=133514392

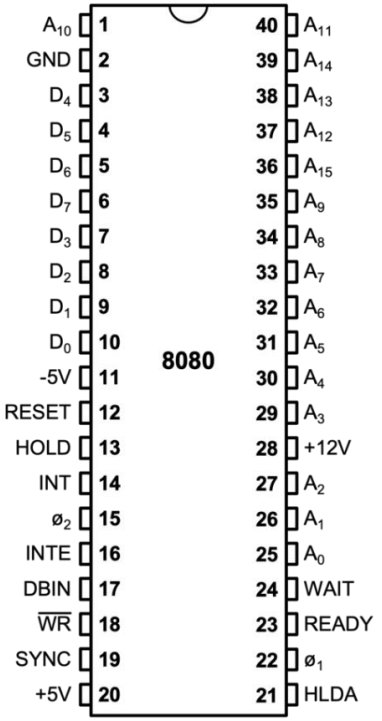
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Intel 8080 Die



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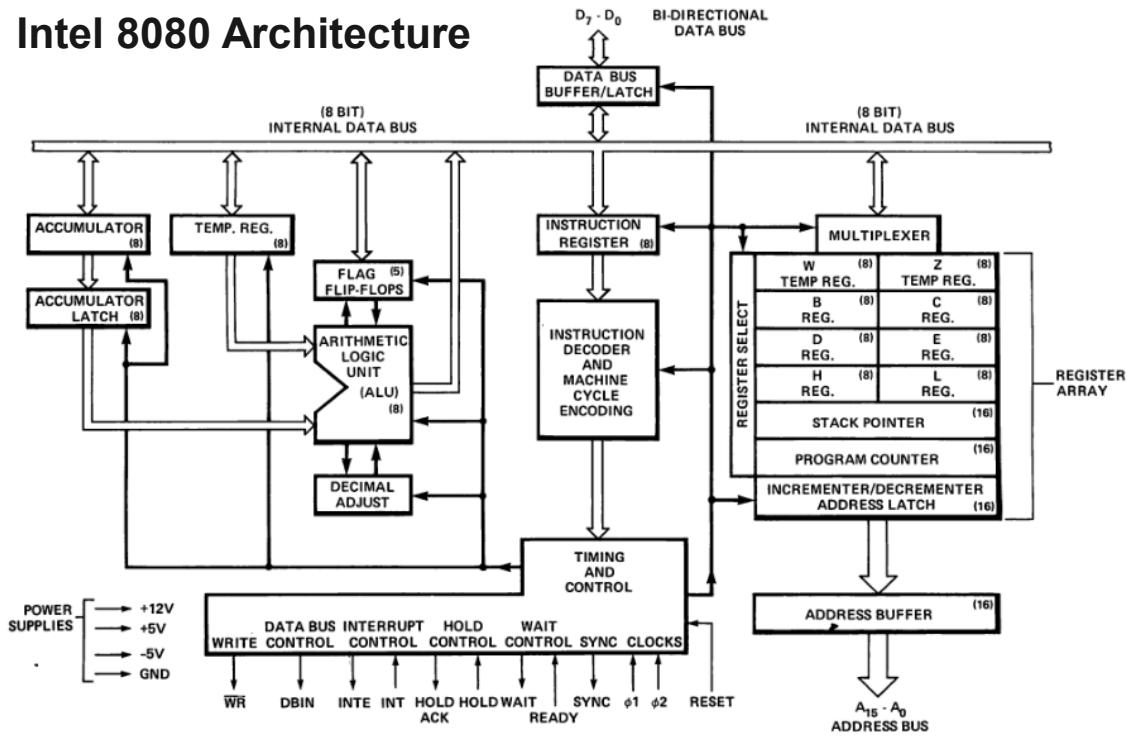
Intel 8080 Pinout



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Intel 8080 Architecture



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Intel 8080 registers

$^{1}_{5}$ $^{1}_{4}$ $^{1}_{3}$ $^{1}_{2}$ $^{1}_{1}$ $^{1}_{0}$ $^{0}_{9}$ $^{0}_{8}$ $^{0}_{7}$ $^{0}_{6}$ $^{0}_{5}$ $^{0}_{4}$ $^{0}_{3}$ $^{0}_{2}$ $^{0}_{1}$ $^{0}_{0}$ *(bit position)*

Main registers

A	Flags	**P**rogram **S**tatus **W**ord
B	C	**B**
D	E	**D**
H	L	**H** (indirect address)

Index registers

SP	**S**tack **P**ointer

Program counter

PC	**P**rogram **C**ounter

Status register

S	Z	0	A_C	0	P	1	C	Flags [13]

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MITS Altair 8800 (1974)



By Swtpc6800 en:User:Swtpc6800 Michael Holley
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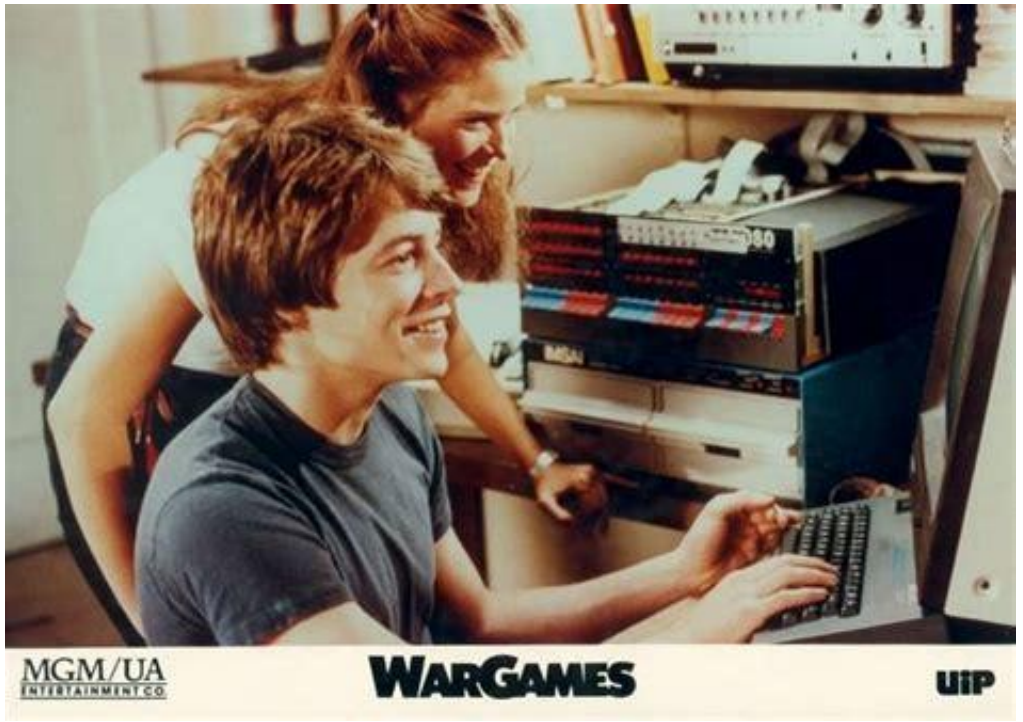
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IMSAI 8080 (1975 to 1978)



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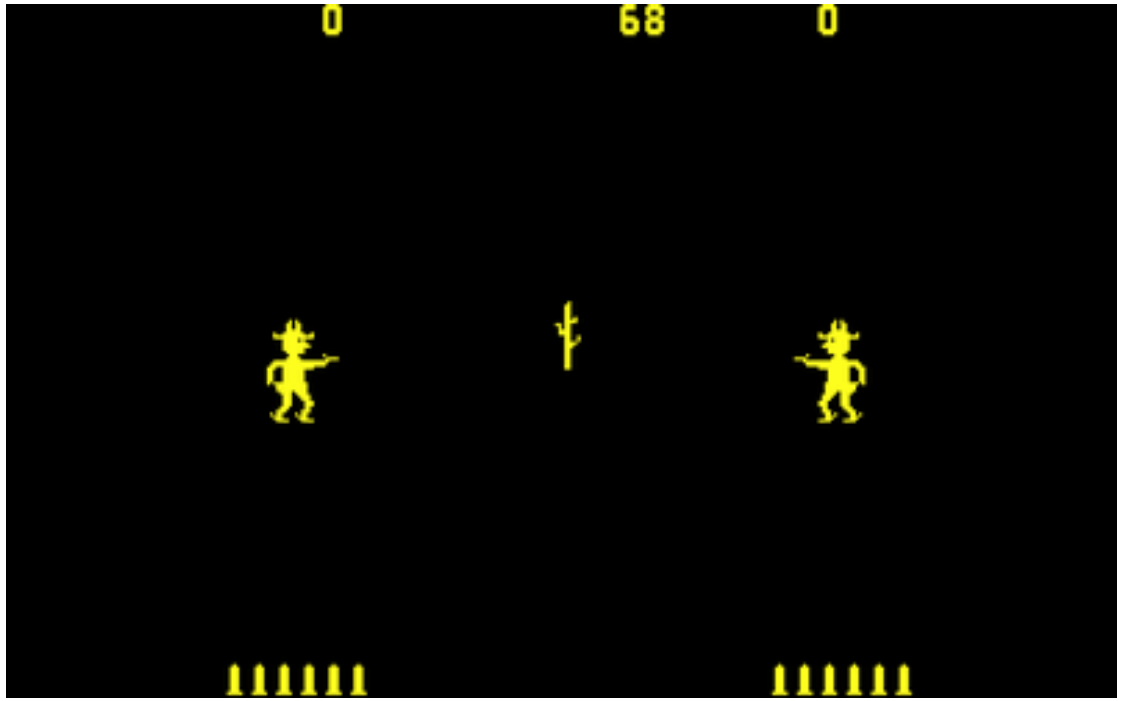
Matthew Broderick and Ally Sheedy (1983)



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Rune's PC-Museum, http://pc-museum.com/046-imsai8080/wargames.htm

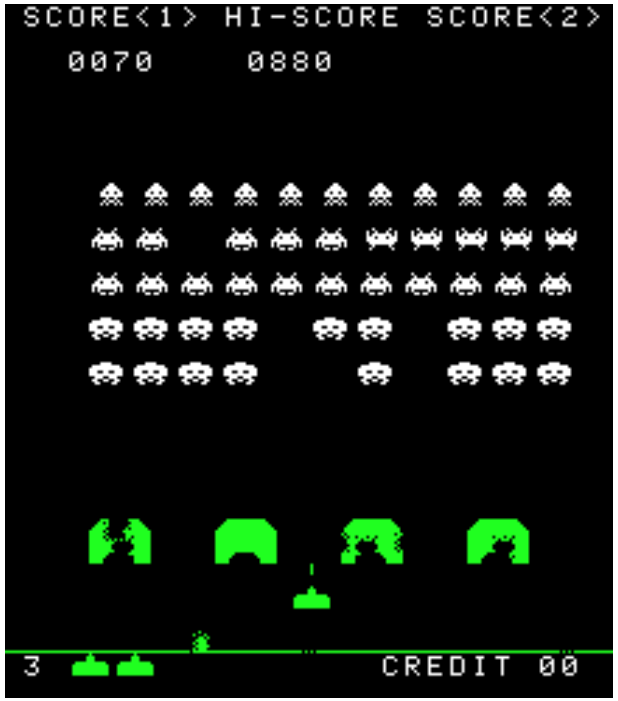
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Gun Fight



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Space Invaders



By The Gamespy Hall of Fame: Space Invaders, Fair use,
https://en.wikipedia.org/w/index.php?curid=17676369

EE426 / 626 Microprocessor Applications

Initial 8080 price: $360 as a nod to the IBM/360

There is an asteroid named “8080 Intel”

Microsoft BASIC was Microsoft’s founding product, fit in to 4kB (with floating point math!)

Microsoft’s published phone number: 425-882-**8080**

Many 8080 core instructions remain in the x86 instruction set architecture (ISA) today

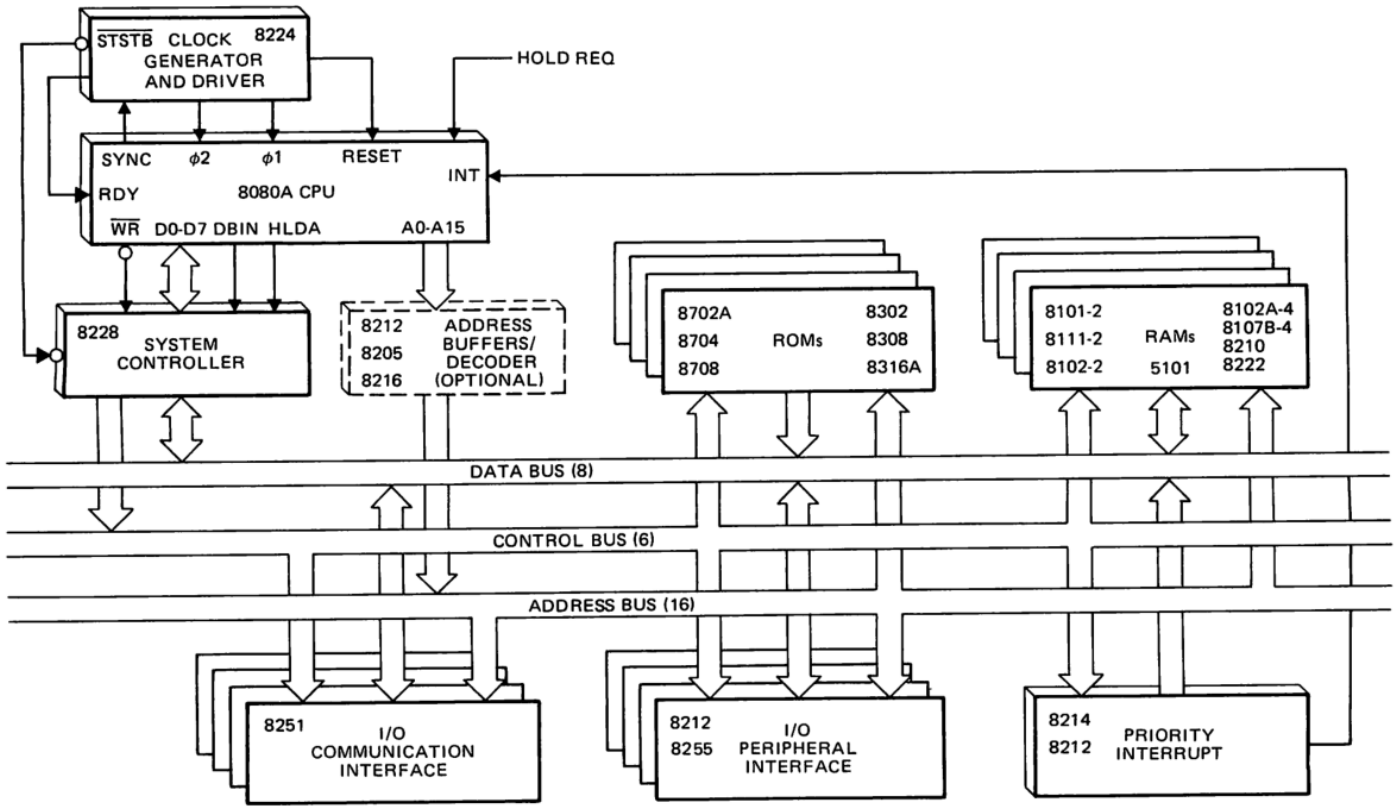
Registers A, B, C, D and control flags continue to exist

PL/M (Programming Language for Microcomputers) developed for 8008 → 8080

CP/M (Control Program for Microcomputers), rejected by Intel ☺

Altair 8800 was named by a 12 year old Star Trek Fan (was the destination in “Amok Time”)

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EE426 / 626 Microprocessor Applications

intel Silicon Gate MOS **8253**

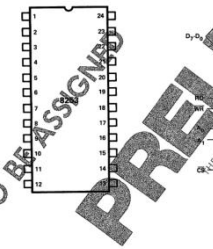
PROGRAMMABLE INTERVAL TIMER

- 3 Independent 16-Bit Counters
- DC to 3 MHz
- Programmable Counter Modes
- Count Binary or BCD
- Single +5V Supply
- 24 Pin Dual-in-line Package

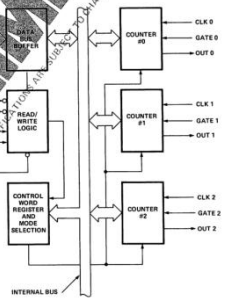
The 8253 is a programmable counter/timer chip designed for use as an 8080 (or 8008) peripheral. It uses nMOS technology with a single +5V supply and is packaged in a 24-pin plastic DIP.

It is organized as three independent 16-bit counters, each with a count rate from 0Hz to 3MHz. All modes of operation are software programmable by the 8080.

PIN CONFIGURATION



BLOCK DIAGRAM



PRELIMINARY
TO BE ASSIGNED
THESE SPECIFICATIONS ARE SUBJECT TO CHANGE WITHOUT NOTICE

5-169

intel Silicon Gate MOS **8255**

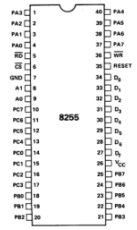
PROGRAMMABLE PERIPHERAL INTERFACE

- 24 Programmable I/O Pins
- Completely TTL Compatible
- Fully Compatible with MCS™ -8 and MCS™-80 Microprocessor Families
- Direct Bit Set/Reset Capability Easing Control Application Interface
- 40 Pin Dual In-Line Package
- Reduces System Package Count

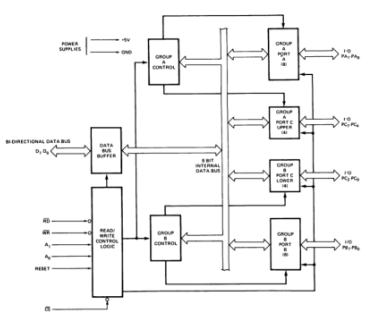
The 8255 is a general purpose programmable I/O device designed for use with both the 8008 and 8080 microprocessors. It has 24 I/O pins which may be individually programmed in two groups of twelve and used in three major modes of operation. In the first mode (Mode 0), each group of twelve I/O pins may be programmed in sets of 4 to be input or output. In Mode 1, the second mode, each group may be programmed to have 8 lines of input or output. Of the remaining four pins three are used for handshaking and interrupt control signals. The third mode of operation (Mode 2) is a Bidirectional Bus mode which uses 8 lines for a bidirectional bus, and five lines, borrowing one from the other group, for handshaking.

Other features of the 8255 include bit set and reset capability and the ability to source 1mA of current at 1.5 volts. This allows darlington transistors to be directly driven for applications such as printers and high voltage displays.

PIN CONFIGURATION



8255 BLOCK DIAGRAM



PIN NAMES

D_0–D_7	DATA BUS (BI DIRECTIONAL)
RESET	RESET INPUT
$\overline{CS}$	CHIP SELECT
$\overline{RD}$	READ INPUT
$\overline{WR}$	WRITE INPUT
A0, A1	PORT ADDRESS
PA7-PA0	PORT A (BIT)
PB7-PB0	PORT B (BIT)
PC7-PC0	PORT C (BIT)
V_{CC}	+5 VOLTS
GND	0 VOLTS

5-113

intel Silicon Gate MOS **8259**

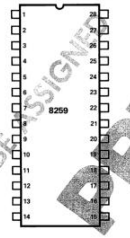
PROGRAMMABLE INTERRUPT CONTROLLER

- Eight Level Priority Controller
- Expandable to 64 Levels
- Programmable Interrupt Modes (Algorithms)
- Individual Request Mask Capability
- Single +5V Supply (No Clocks)
- 28 Pin Dual-in-Line Package

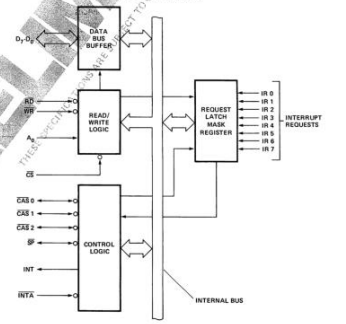
The 8259 handles up to eight vectored priority interrupts for the 8080A CPU. It is cascadable for up to 64 vectored priority interrupts, without additional circuitry. It will be packaged in a 28-pin plastic DIP, uses nMOS technology and requires a single +5V supply. Circuitry is static, requiring no clock input.

The 8259 is designed to minimize the software and real time overhead in handling multi-level priority interrupts. It has several modes, permitting optimization for a variety of system requirements.

PIN CONFIGURATION



BLOCK DIAGRAM



PRELIMINARY
TO BE ASSIGNED

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Fair use, Intel 8080 Microcomputer Systems User's Manual, September 1975